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24 October 1986
SUPERSEDING
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(See 6.7)

MILITARY SPECIFICATION

LIFE PRESERVERS, CO_2 INFLATABLE, POUCH TYPE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers pouch type CO_2 inflatable life preservers in the colors specified in 3.3.5.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton.
V-T-295 - Thread, Nylon.
QQ-A-250/8 - Aluminum Alloy 5052, Plate and Sheet.
QQ-B-613 - Brass, Leaded and Non-leaded; Flat Products (Plate, Bar, Sheet, and Strip).
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-W-343 - Wire, Electrical, Copper (Uninsulated).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-F-495 - Finish, Chemical, Black, for Copper Alloys.
MIL-W-530 - Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-R-900 - Rubber Gasket Material, 45 Durometer Hardness.
MIL-L-3454 - Life Preservers; Packaging of.
MIL-A-5540 - Adhesive, Polychloroprene.
MIL-W-17337 - Webbing, Textile, Woven Nylon.
MIL-C-17415 - Cloth, Coated, and Webbing, Inflatable Boat and Miscellaneous Use.
MIL-C-19002 - Cloth, Coated, and Strip, Coated Cloth - Polychloroprene on Nylon.
MIL-C-19377 - Cloth, Twill, Nylon.
MIL-C-21768 - Copper Alloy Numbers 210 (Gliding, 95%) and 220 (Commercial Bronze, 90%) Sheet and Strip.
MIL-C-25369 - Cylinders, Carbon Dioxide, for Inflation of Pneumatic Life Preservers.
MIL-P-25732 - Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Limited Service at 275°F (135°C).
MIL-T-43566 - Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors.
MIL-O-81375 - Oral Inflation Assemblies, Survival Equipment, Inflatable.

STANDARDS

FEDERAL

FED-STD-595 - Colors.
FED-STD-601 - Rubber; Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

AIR FORCE-NAVY AERONAUTICAL BULLETIN AN-227 - Fasteners - Snap and Curtain.

BUREAU OF SHIPS

803-1183395 - Rubber Sleeve Check Valve for CO_2 Inflatable Life Preserver.
803-1230582 - Life Preserver, CO_2 Inflatable, With Pouch: Pouch.
803-1419244 - Life Preserver, CO_2 Inflatable, With Pouch: Webbing Harness and Details.
803-1419350 - Life Preserver, CO_2 Inflatable, With Pouch: Assembly.
803-1419351 - Life Preserver, CO_2 Inflatable, With Pouch: General Arrangement.

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BUREAU OF SHIPS (Continued)

803-1419352 - Life Preserver, CO_2 Inflatable, With Pouch: Assembly of Details.

803-1419353 - Life Preserver, CO_2 Inflatable, With Pouch: Development of Details.

803-1411524 - CO_2 Holder and Holder Gauge Details.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 117 - Standard Method of Salt Spray (Fog) Testing. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race, Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Life preservers furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Material. In instances where the drawings referenced herein do not agree with the materials or design specified in this specification, this specification shall govern.

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3.3.1 Coated cloth. The coated cloth used in the construction of the life preserver shall conform to type I of MIL-C-19002. The coated cloth tape used in the construction of the life preserver shall conform to type III of MIL-C-19002. The coated cloth used in the construction of the pouch shall conform to type 2, class A of MIL-C-17415, except that the requirement for hydrostatic resistance and resistance to accelerated weathering shall not apply. The coated cloth and tape shall be of a quality that will withstand the performance requirements specified herein when fabricated into a life preserver.

3.3.2 Webbing.

3.3.2.1 Cotton webbing. The dyed cotton webbing used in the life preserver shall be of the following widths and types:

Part	Width (inches)	Type
Webbing harness:		
Belt	2	III - Heavyweight of MIL-W-530
Retaining strap	3/4	I - Lightweight of MIL-T-43566

3.3.2.1.1 The cotton webbing shall be made mildew-resistant and water-repellent in accordance with MIL-W-530.

3.3.2.2 Nylon webbing. The dyed nylon webbing used in the life preservers shall conform to MIL-W-17337. The webbing shall be 1 inch wide for the hoisting strap and the buckle strap.

3.3.3 *Mechanical inflation system*.

3.3.3.1 Holder assemblies. Each life preserver shall be provided with one holder assembly conforming to the dimensions shown on Drawing 803-1411524. The assembly shall consist of a holder designed for use with a CO_2 cylinder conforming to MIL-C-25369 and shall have an external actuating lever with lanyard, internal piercing pin, and coin set screw. The holder shall be a high pressure aluminum die casting and shall be anodized and treated with dichromate seal. The holder shall be fitted with an O-ring gasket as shown on Drawing 803-1419352. The gasket shall conform to MIL-P-25732. The design shall be such that the cylinder will be punctured and the piercing pin will be retracted with a travel of the actuating lever through an arc of not over 90 degrees. No side pull shall be required to puncture the cylinder and retract the pin. The discharge mechanism shall operate on the eccentric principle, which will allow the piercing point to return to its original position after operation even when the lever is held in the tripped position. When not is use, there shall be provision for holding the actuating lever against the side of the holder by a length of 0.016-inch diameter safety wire conforming to type 5 of QQ-W-343. The lever shall be corrosion-resisting hardened steel having a minimum Rockwell 45-N hardness of 40. The lever shall be held in place with a staked corrosion-resisting steel pin. The coin set-screw shall be steel, corrosion-resistant, and sodium-dichromate dipped. The holder threads shall be coated with a silicone grease after the holder has been assembled on the life preserver. There shall be a corrosion-resisting music wire or corrosion-resisting steel spring to provide that the

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piercing pin separates from the cylinder immediately after puncturing to preclude any possibility of delay in inflation by the pin remaining in and sealing the hole. The piercing pin shall be a corrosion-resisting hardened steel having a minimum Rockwell 45-N hardness of 50. The piercing point of the pin shall conform to the design shown on Drawing 803-1411524. The piercing pin shall be so sealed that there shall be no leakage of gas from the holder when the discharge lever is in any position. An O-ring may be provided on the shank of the piercing pin to accomplish this. The piercing pin in its normal position shall be at least 0.010 inch behind the cylinder seat to prevent accidental puncture. The holder shall be so designed that adequate clearance is provided between the open end of the holder and the CO_2 cylinder. There shall be adequate clearance so that if the surface where the puncturable end of the cylinder seats within the holder becomes worn, the cylinder can be screwed into the holder to a greater depth and onto its seat. Holders shall withstand a hydrostatic pressure of 3500 pounds per square inch (lb/in^2). The holder shall show no change in puncture ability when tested as specified in 4.9.11. The design of the entire inflation system shall be submitted to the command or agency concerned for tentative approval.

3.3.3.2 Holder lanyard. The holder lanyard shall consist of a 1/8-inch diameter commercial braided nylon cord, a plastic button, and two black nickel-plated brass clips. When attached to the actuating lever, the cord shall be doubled back upon itself to form a loop. The two free ends of the cord shall be inserted through the holes in the bottom of the pull cord button and then tied in a sheet bend knot. A cord slip shall be attached to the cord adjacent to the pull cord button and adjacent to the discharge lever to hold the double cord together. The entire assembly of cord, clips, and button shall not exceed 2-1/2 inches in overall length. The discharge lever, cord, and button shall withstand a minimum pull of 100 pounds.

3.3.3.3 Manifold. The holder shall be securely attached to the life preserver by means of a brass manifold molded to a properly compounded neoprene base which shall in turn be cemented to the inside surface of the life preserver fabric so that the holder projects on the outside. The brass manifold, including threads, shall be coated with electrodeposited nickel conforming to class 1, grade E of QQ-N-290. The manifold and holder shall be equipped with flats to maintain the holder in a fixed position. The manifold shall conform to piece number 38 of Drawing 803-1419352 and shall be properly designed to fit the holder. The manifold shall be located and oriented as shown on Drawing 803-1419353. The manifold shall be provided with a one-way check valve conforming to Drawing 803-1183395. The holder and manifold shall be assembled as shown on Drawing 803-1419352. Two synthetic rubber gaskets conforming to MIL-R-900 shall be provided, one for each side of the holder attachment. The hexagonal nut shall be nickel-plated brass.

3.3.4 Oral inflation valve. The oral inflation valve shall conform to type I of MIL-O-81375, except that the weight shall be 30.0 grams (maximum), and the valve shall withstand without damage inflation by means of a blow gun having a discharge nozzle not exceeding 1/8 inch diameter when the gun is connected to a pressure source of 225 lb/in^2.

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3.3.5 Color.

3.3.5.1 Unless otherwise specified in the contract or order (see 6.2.1), the color of the life preserver cloth shall acceptably match color chip number 12246 of FED-STD-595. The color of the pouch, belt, retaining straps, hoisting strap and buckle strap shall acceptably match color chip number 26008 of FED-STD-595.

3.3.5.2 When specified (see 6.2.1), the color of the life preserver cloth shall match color chip number 26008 of FED-STD-595 or the color of the life preserver cloth and the pouch, belt, retaining straps, hoisting strap, and buckle strap shall match color chip number 14064 of FED-STD-595.

3.3.6 Thread. Thread used for sewing the pouch and nylon webbing shall conform to type II, size E of V-T-295, colored to match the pouch and webbing. Thread used for sewing the cotton webbing shall be type 1B3, number 16, four-ply cotton thread equal in color to the webbing and conforming to the physical characteristics and color fastness requirements of V-T-276.

3.3.7 Harness hardware. Metal harness hardware shall conform to Drawing 803-1419244. All metal harness hardware, except buckle and ball type tip, shall be bronze stampings at least equal to the requirements of MIL-C-21786. The ball type tip shall be brass at least equal to the requirements of QQ-B-613. The buckle shall be an aluminum stamping at least equal to the requirements of QQ-A-250/8. All harness hardware except the buckle shall have a lusterless black finish conforming to MIL-F-495. The buckle shall have an anodized finish.

3.3.8 Adhesive. The adhesive used shall be polychloroprene conforming to class 1, 2, or 3 of MIL-A-5540, except that qualification tests are not applicable, the color shall acceptably match the life preserver cloth, and the life preserver coated cloth shall be substituted for the cotton duck for the adhesion tests.

3.3.9 Reflective tape. The material for the reflective tape for application to inflatable life preservers shall be number 3150, SOLAS grade, reflective material, 2 inches wide, Minnesota Mining and Manufacturing Company or equivalent. The primer for preparing surfaces prior to cementing shall be Minnesota Mining and Manufacturing Company primer number 2141 or equivalent. Reflective tape shall be located and cemented as shown on Drawing 803-1419350 and as specified in 3.4.8.

3.4 Manufacture and assembly.

3.4.1 Details. The manufacture and assembly details of the life preservers shall conform to Drawings 803-1183395, 803-1230582, 803-1419244, 803-1419350, 803-1419351, 803-1419352, 803-1419353, and 803-1411524.

3.4.2 Pouch. The pouch shall conform to Drawing 803-1230582 and shall be made of coated cloth (see 3.3.1).

3.4.3 Toggle line. The toggle line as shown on Drawings 803-1419350 and 803-1419244 shall be 1/8 inch diameter commercial braided nylon cord.

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3.4.4 Oral inflation. The oral inflation valve shall be installed on the outside of the life preserver and shall communicate with the air compartment by means of a properly compounded neoprene connector 3/8-inch (minimum) inside diameter, 9/16-inch outside diameter. The valve shall be secured to the connector by a suitable clamp to withstand a load of 50 pounds.

3.4.5 Snap fasteners. The snap fasteners used for the CO_2 inflator cover shall be attached as shown in detail A of Drawing 803-1419353 and shall be located as shown on Drawing 803-1410350. Three sets of fasteners shall be furnished. Snap fasteners shall be in accordance with Drawing AN-227, 49B, 50B, 51B, and 52B and shall have a black finish.

3.4.6 Press stud. Press stud fasteners conforming to Drawing AN-227, 6B, 7B, 8B, and 9B shall be used for the pouch (see Drawing 803-1230582), the harness (see Drawing 803-1419244), and the buoyancy chamber panel (see Drawing 803-1419351). The method of attaching the press studs at the buoyancy chamber panel shall be as shown in assembly of press stud patch on Drawing 803-1419352.

3.4.7 Light attachment loop, inflation tube loop, and CO_2 cylinder loop. The light attachment loop, inflation tube loop, and CO_2 cylinder loop shall be as shown on Drawing 803-1419353 and shall be made of two plies of coated cloth (see 3.3.1).

3.4.8 Cementing.

3.4.8.1 In the construction of the seams, sealing area, and attachment of all component parts of the life preserver, either cementing or pressure-curing by heat shall be used. In all cementing operations, the surfaces to which the adhesive is to be applied shall be thoroughly washed, if necessary, with a suitable solvent in such a manner that the dusting materials (zinc stearate or talc) and any other surface contaminant is completely removed. The surface shall be cleaned prior to cementing. Care shall be exercised to ensure that neither the coating nor the base undercloth is damaged, nor that the adhesion between both is impaired in any way by prolonged exposure to the solvent. The solvent used shall evaporate completely prior to the application of the adhesive and shall leave no residue. Cemented areas shall be rolled and stitched out, and all trapped air and wrinkles that would impair the function of the preserver shall be removed. Cements shall be controlled to ensure that old cement or cement which has partly set up is not used. A fresh batch of cement shall be mixed at least every 8 hours during the course of manufacture. Containers for cement shall be free from congealed cement before being refilled.

3.4.8.2 Cemented area. The cement on all outside cemented areas in way of taped seams and patches shall be applied to the area and shall extend not less than 1/8 inch and not more than 3/8 inch beyond the edges of the tape or patch and parallel thereto.

3.4.8.3 Cementing seams and sealing areas. The construction of seams and sealing areas and the attachment of all component parts to the preserver shall be undertaken utilizing the technique and precautions specified in 3.4.8.1 so that all such seams and sealed areas shall develop the optimum properties of the adhesive as specified in MIL-A-5540 before submission of the completed preserver for final approval.

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3.4.9 Marking. Identification marking shall be by the silk screen method in black letters and numerals 1/4 inch high. A marking patch shall be located on the outside front of the life preserver as shown on Drawing 803-1419351 and the outside surface of the pouch flap shall be marked as shown on Drawing 803-1230582. The date of manufacture and serial number may be rubber stamped.

3.4.9.1 Serial numbers. The successful bidder will be furnished with applicable serial numbers at the time of award of the contract or order.

3.5 Workmanship. The workmanship shall be in accordance with high grade manufacturing practice covering this class of work. Because of the emergency use of this equipment, it is required that all details of workmanship be of the highest quality in order to ensure proper functioning of the preservers at all times.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test reports. When specified in the contract or order, a test report shall be prepared (see 6.2.2).

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection of packaging (see 4.11).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). The inspection performed on the completed life preservers and component material shall consist of all the tests specified in this specification. Samples of all materials used in the fabrication of the life preserver shall be examined and tested to ascertain their conformance to the applicable specifications or to the requirements for the material as specified herein.

4.3.1 Samples for qualification inspection. The following samples shall be submitted for qualification inspection:

(a) Two complete life preservers.
(b) Coated fabric samples designated in MIL-C-19002 together with the name and code number of the fabric supplier.
(c) Uncoated cloth samples as designated in MIL-C-19377, together with the name and code number of the cloth supplier.
(d) Three square yards of coated and uncoated cloth used in the pouch.
(e) Two square feet of cured pouch coating compound (normal 6 by 6 inch squares) 0.07 to 0.08 inch thick.
(f) Two coating adhesion samples 12 inches wide and 12 inches long prepared as specified in MIL-C-17415.
(g) Five yards of cotton webbing of each width and type used in the life preserver.
(h) Five yards of nylon webbing of the width and type used in the life preserver, together with the name and code number of the webbing supplier.
(i) One cone (500 yards) of nylon sewing thread and one cone of cotton sewing thread of each color used in the life preserver.
(j) Four sets of all hardware, including the oral and CO_2 inflation systems, used in life preservers, together with the name and code number of the hardware supplier.
(k) Five yards of safety wire.

4.3.1.1 Qualification inspection samples shall be plainly identified by securely attached durable tags marked with the following information:

(a) "Samples for qualification inspection".
(b) Name of product (together with complete identifying data, such as manufacturer's designation or number).
(c) Name of manufacturer.
(d) Submitted by (name), (date) for qualification inspection in accordance with the requirements of MIL-L-15581 under authorization (reference authorizing letter).

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4.4 First article inspection. Before starting regular production, the contractor shall submit samples as specified in 4.3.1.1. These samples shall be subjected to the examination and tests of 4.6 and 4.9.

4.4.1 First article test samples shall be plainly identified by securely attached durable tags marked with the following information:

(a) "Samples for first article test."
(b) Name of product, together with complete identifying data, such as manufacturer's designation number.
(c) Name of manufacturer.
(d) Contract or order number.

4.5 Quality conformance inspection. Quality conformance inspection shall be performed by the contractor under the surveillance of the Government inspector. The inspection shall consist of the examination specified in 4.6 and the tests specified in 4.5.1.3 and 4.5.1.4. The buoyancy test specified in 4.9.4 shall be performed when the CO_2 cylinder is expended during the operation test specified in 4.9.1. The results of the pressure, leakage, and buoyancy tests shall be identifiable by listing the assigned serial numbers. To ensure that the final product is sound and that it conforms to the requirements of this specification, the contractor shall provide for the adequate control of the quality of the coated fabric, the adhesive, and the attachments during the process of manufacture.

4.5.1 Sampling for quality conformance inspection.

4.5.1.1 Lot. A lot shall consist of all life preservers submitted for inspection at one time.

4.5.1.2 Sampling for examination. A random sample of life preservers shall be selected from each lot in accordance with inspection level I of MIL-STD-105 for the examination specified in 4.6. The acceptance number shall be zero for critical defects. The acceptable quality level (AQL) shall be 4.0 for major defects and 10.0 for minor defects per 100 units.

4.5.1.3 Sampling for tests. A random sample of life preservers shall be selected from each lot in accordance with the following inspection levels as specified in MIL-STD-105 for the tests specified:

Tests	Inspection level
Operation (see 4.9.1)	I
Pressure (see 4.9.2)	I
Leakage (see 4.9.3)	I
Buoyancy (see 4.9.4)	S-3

Life preservers not meeting the requirements of these tests may be corrected before submittal for acceptance, but under no circumstances shall an air retaining seam be reworked more than one time. The acceptance number for all tests shall be zero. Rejected life preservers shall not be resubmitted.

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4.5.1.4 Additional tests. The tests shown in table I shall be conducted either by the contractor or by the subcontractor at the point of manufacture. In addition, the contractor shall prepare a minimum of ten adhesion test samples from each day's production and shall subject the samples to the tests specified in 4.9.8. The dimensions of the test specimen shall be in accordance with the requirements for strip specimens for the test in accordance with method number 8011 as specified in FED-STD-601.

TABLE I. Quality conformance tests.

Tests	Paragraph	Sample plan
Leakage and operation (holder assembly)		
40 lb/in^2 test	4.9.5	100 percent
3500 lb/in^2 hydrostatic pressure test 1/	4.9.5	1 sample from each 500
Oral inflation tube (porosity)	4.9.6.1	100 percent
Oral inflation tube (tensile)	4.9.6.2	Six samples from each 1000
Dimensional inspection (gauge) of holder assembly	4.9.7	100 percent
Oral inflation valve adhesion	4.9.10	Six samples from each 1000
Holder efficiency	4.9.11	25 samples from each 500

1/ Failure of any holder subjected to the hydrostatic pressure test shall be cause for rejection of the entire lot from which the sample was selected.

4.6 Examination. Sample life preservers selected in accordance with 4.5.1.2 shall be thoroughly examined to determine conformance to the requirements of this specification and the applicable drawings. Examination shall be conducted as specified in table II.

TABLE II. Classification of defects.

Categories	Defects
Critical:	
1	Inflatable compartment material defective or not as specified; evidence of splits, patched holes, tears, fraying.
2	Inflatable compartment seams defective, incomplete, or not as specified; evidence of bubbles, improperly applied tape, incomplete cementing, or seams in inflatable compartment not the specified type; width of seams less than the required minimum.
3	Manifold or oral inflation tube defective, not secure, securing patches not as required, or diameter of base not as specified.
4	Oral inflation valve defective or not secure in tube; tube evidences cracks, splits, bubbles, bulges, pin holes, or other deleterious defects.
5	Holder assembly defective, not securely attached, or not as specified; lanyard not securely attached to actuating lever; lever holding pin not staked in place.
6	Failure of specified gauge to seat firmly against seating surface of holder; O-rings for cylinder and piercing pin missing, defective, or not as specified (examine prior to assembly).

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TABLE II. Classification of defects. - Continued

Categories	Defects
Major:	
101	Loops for securing CO_2 cylinder harness or inflation tube defective or not securely attached as specified.
102	Holder and marking patch not secure, inflation cover not secured to holder and marking patch, or snap fasteners missing, defective, or not secured to patches as required.
103	Pouch defective, not securely attached as required, or stitching incomplete; snap fasteners defective or not secured as required.
104	Belt or harness webbing evidences tears, splits, fraying, or weak spots.
105	Belt or harness stitching defective, incomplete, or not as specified.
106	Belt or hoisting strap hardware missing, defective, or not secure; evidence of cracks, deep pits, thin spots, or sharp surfaces; buckle cannot be engaged, or length of belt or strap not adjustable.
107	Belt or hoisting strap length not as required.
108	Retaining straps missing or defective; snap fasteners defective or not secure.
109	Toggle line defective, not knotted to belt, or toggle not secure.
110	Holder actuating lever not safety wired.
111	Metal components (belt hardware, holder, lever, or manifold) not plated or finished as specified; plating or finish defective or incomplete.
112	Date of manufacture marking missing, illegible, or incorrect.
113	Set screw provided in holder for holding CO_2 cylinder missing or not staked.
114	Light attachment loop defective or not attached as specified.
115	Reflective tape shows evidence of tears or splits; tape not cemented as specified.
Minor:	
201	Hoisting loop dimension not as required.
202	Length of lanyard or toggle line not as required.
203	Marking, manufacturer's name, or trademark missing, illegible, incorrect, or not as specified.
204	Cement on fabric surface around accessory patches and tapes less than 1/8 inch or greater than 3/8 inch.

4.6.1 Resubmitted lots. Resubmitted lots (allowed in connection with major and minor defects found in the examination) may be resubmitted for examination as specified in 4.5.1.2. Where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be selected with the acceptance number remaining zero. The resubmitted lot shall be examined by the contractor under the surveillance of the Government inspector using tightened examination. Before a lot is resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the lot shall be furnished by the contractor to the Government inspector.

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4.7 Comparison tests. NAVSEA may conduct at a Government laboratory comparison tests on random samples of life preservers produced under any contract. Comparison tests may include any or all of the tests of this specification. Acceptance and shipment of life preservers shall not be delayed pending completion of comparison tests. However, if any comparison test sample is rejected for noncompliance with the requirements of this specification, acceptance and shipment of any further life preserver shall be withheld until the contractor can demonstrate that he has corrected the conditions which led to the rejection of the comparison test sample. Samples selected for comparison tests shall be considered a part of the quantity to be delivered under the contract and shall be returned with all applicable test reports, recommendations, and comments pertinent for monitoring production.

4.8 Test conditions.

4.8.1 Atmospheric conditions. Unless otherwise specified in the contract or order, all tests required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury (Hg) and a temperature of 77 $\pm$ 18 degrees Fahrenheit (°F) and a relative humidity of 80 percent or less. If the final values of the ambient temperature or barometric pressure at the end of the 24 hour air pressure tests are different from the initial values, the following corrections shall be made to the final pressure readings:

(a) Temperature corrections: For each rise in temperature, 0.031 lb/in^2 shall be substracted from the final pressure reading. For each degree drop in temperature, 0.031 lb/in^2 shall be added to the final pressure reading.
(b) Barometric pressure correction: For each 0.1 inch of Hg rise in barometric pressure, 0.049 lb/in^2 shall be added to the final temperature corrected pressure reading. For each 0.1 inch of Hg drop in barometric pressure, 0.049 lb/in^2 shall be subtracted from the final temperature corrected pressure reading.
(c) Pressure shall be measured by means of a Hg manometer and then converted to lb/in^2 by multiplying inches of Hg by 0.49.

4.8.2 CO_2 cylinder. The CO_2 cylinder used in conducting the tests specified in 4.9.1 and 4.9.4 shall conform to MIL-C-25369 except that the gas charge shall be not less than 17 grams nor more than 24 grams of CO_2. The cylinder shall be marked with a permanent type ink as follows: "For Test Purposes Only" and the inclusion of a nonapplicable contract number or the omission of the contract number is acceptable.

4.9 Test methods.

4.9.1 Operation. The operation of the preserver shall be tested by releasing a CO_2 cylinder into the life preserver in the normal manner. The preserver shall inflate within 5 seconds from the time the CO_2 cylinder is discharged. Remove the empty cylinder from the holder after discharge and allow the preserver to remain on test for 30 minutes before performing the pressure test specified in 4.9.2.

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4.9.2 Pressure. The life preserver shall be inflated with air through the oral inflation valve to a pressure of 5 lb/in^2 and shall be allowed to stand for a minimum of 5 minutes. The pressure drop shall not be greater than 0.4 lb/in^2. The life preserver shall be examined while held under pressure. There shall be no evidence of failure of the materials in any respect. All seams and sealed areas shall remain perfectly intact and shall show no signs of separation.

4.9.3 Leakage. The life preserver shall be inflated with air through the oral inflation valve to a pressure of 2 lb/in^2 and shall be allowed to stand for a minimum of 15 minutes. The pressure shall then be checked and readjusted if necessary to the original pressure of 2 lb/in^2. The temperature and barometric pressure shall be recorded at this time. At the end of a minimum of 24 hours after the readjustment period the pressure shall be measured and corrected for any change in temperature or barometric pressure (see 4.8). The corrected pressure shall be not less than 1.5 lb/in^2. The oral inflation valve shall be examined while the life preserver is being deflated to determine that the valve works without difficulty and shuts off securely.

4.9.4 Buoyancy. Immediately upon completion of the operation test (see 4.9.1), with the oral inflation valve locked shut, the life preserver shall be attached to a steel weight, the latter weighing 29 pounds when submerged in fresh clean water. The attached weight shall not result in the complete submergence of the life preserver.

4.9.5 Leakage and operation (holder assembly). Holders shall be tested for leakage under water with air at a pressure of 40 lb/in^2 with the actuating lever in the actuated position. (The actuated position is considered to be the position of the lever after the pin has punctured the cylinder and has completely retracted. Cylinders are not required to determine the actuated position.) Holders shall withstand a hydrostatic pressure of 3500 lb/in^2. Holders shall also be tested for leakage under water with the actuating lever slowly carried through one complete cycle.

4.9.6 Oral inflation tube. Before the oral inflation valve is clamped in the oral inflation tube, the tests specified in 4.9.6.1 and 4.9.6.2 shall be conducted.

4.9.6.1 Porosity. The oral inflation tube shall be subjected to an air pressure of 25 lb/in^2 for a minimum of 10 seconds while submerged under water. There shall be no evidence of air leakage due to porosity. There shall be no evidence of weakening of the tube as revealed by excessive deformation in any portion of the tube.

4.9.6.2 Tensile. The oral inflation tube base shall be folded longitudinally so that the rounded edges coincide and then placed in the lower jaws of a suitable test machine. The tube portion shall be placed in the upper jaws. The rate of load application shall be 12 $\pm$ 1/2 inches per minute (in/min). The tensile strength shall be not less than 75 pounds.

4.9.7 Dimensional inspection (gauge) of holder assembly. The holder assembly, complete with O-ring, shall be dimensionally checked with a plug gauge conforming to the dimensions shown on Drawing 803-1411524. Failure of the plug gauge to seat firmly against the seating surface (where the puncturable end of the cylinder seats within the holder) shall be cause for rejection.

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4.9.8 Adhesion. The test for adhesion shall be conducted in accordance with method number 8011 specified in FED-STD-601. The adhesive strength of the cemented samples shall be not less than 7.5 lb/in of width for samples joining the coated sides of the finished fabric and not less than 5.0 lb/in of width for seams joining coated side to uncoated side of the finished fabric.

4.9.9 Salt spray. A 100-hour salt spray test shall be conducted on the CO_2 inflation devices, oral inflation valve, and all metal parts in accordance with ASTM B 117. The metal fittings shall show no appreciable corrosion or other damage at the end of this test.

4.9.10 Oral inflation valve adhesion. This test shall be conducted after the oral inflation valve is clamped within the oral inflation tube. The oral inflation tube base shall be folded longitudinally so that the rounded edges coincide and then placed in the lower jaws of a suitable machine. The oral inflation valve shall be placed in the upper jaws. The load application shall be 12 $\pm$ 1/2 in/min. The rate to separate the oral inflation valve from the oral inflation tube shall be not less than 50 pounds.

4.9.11 Holder efficiency. The amount of force required to actuate the holder and discharge the cylinder contents shall not exceed 17 pounds at ambient temperature when measured on a tensile machine operating at a speed of 12 $\pm$ 1 in/min with the direction of pull on the holder lever arm in a plane parallel to the longitudinal axis of the holder. Each holder in the sample (see table I) shall withstand one CO_2 firing. In addition, three holders shall be selected from each sample. Each of the three holders shall withstand 15 repeated firings of empty-capped cylinders without change in puncture ability.

4.10 Possible test failures. Possible test failures shall be determined as follows:

(a) Operation: Failure of life preserver to inflate when lanyard is pulled; evidence of leaks through fabric, seams, holder, piercing pin O-rings, or manifold check valve after inflation. Life preserver cannot be readily inflated through the oral inflation valve or evidence of leakage back through the valve after inflation.
(b) Displacement: Displacement of life preserver not within the required maximum and minimum values when inflated and measured as specified.
(c) Leakage: Displacement of life preserver less than the minimum permissible after it has been inflated and then allowed to stand for the required length of time.
(d) Pressure test: Evidence of material failure or parting of seams when life preserver is subjected to the specified pressure.
(e) Buoyancy: Failure of inflated life preserver to support, without sinking, the specified weight.

4.11 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Packaging, packing and marking (see 6.2.1). Life preservers shall be packaged level A or C; packed level A, B, or C; and marked as specified in accordance with MIL-P-3454.

6. NOTES

6.1 Intended use. Life preservers made in accordance with this specification are intended for use on surface ships.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When color is other than as specified (see 3.3.5.1 and 3.3.5.2).
(c) Selection of applicable levels of packaging, packing and marking (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.2	Reports, test	DI-T-2072	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-15581 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of the samples specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Cementing
Coated cloth
Cotton webbing
Nylon webbing

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-15581K(SH)	LIFE PRESERVERS, CO_2 INFLATABLE, POUCH TYPE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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